

			$= 6.2 \times 10^{-10} \text{ N}$	A1
6	(a)		Let J be the contact point between the jockey and the wire XY $R_{XJ} = \frac{L_{XJ}}{L_{XY}} R_{XY} = \frac{0.48}{1.2} (20.0) = 8.0 \Omega$ $V_{XJ} = \frac{R_{XJ}}{R_{XY}} (4.0) = \frac{8.0}{23.0} (4.0) = 1.3913 \text{ V}$ By considering the secondary circuit, p.d. across resistor R = 1.3913 V p.d. across 1.0Ω resistor = 0.50 V By principle of potential divider, $\frac{R_{1\Omega}}{R_{\text{resistor R}}} = \frac{V_{1\Omega}}{V_{\text{resistor R}}}$ $R_{\text{resistor R}} = \frac{1.3913}{0.50} (1.0) = 2.8 \Omega$	C1 C1 A1
	(b)	(i)	$I = nAvq$ $v = I / nAq = 2.8 / ((8.49 \times 10^{28})(0.20 \times 10^{-6})(1.6 \times 10^{-19}))$ $= 1.03 \times 10^{-3} \text{ m s}^{-1}$	M1 A1
		(ii)	$R = \frac{V}{I} = \frac{0.72}{2.8}$ $R = \frac{\rho l}{A}$ $\rho = \frac{RA}{l} = \frac{\left(\frac{0.72}{2.8}\right)(0.20 \times 10^{-6})}{3.0} = 1.7 \times 10^{-8} \Omega \text{ m}$	C1 A1
		(iii)	As the current increases, the temperature increases since the <u>rate at which electrons collide into the metal lattice increases</u>. As temperature increases, <u>lattice ions vibrate more vigorously, hindering the flow of electrons</u>. Hence, its resistance increases.	B1 B1
7	(c)		A discrete unit/quantum/packet of electromagnetic radiation	A1